

Biology - Undergraduate Level Assessment 2

Instructions

- Answer all questions.

Multiple Choice

Question 1

(2 marks)

In most ecosystems, net primary productivity is important because it represents the

- A. biomass of all producers
- B. biomass of all consumers
- C. energy available to heterotrophs
- D. energy consumed by decomposers
- E. total solar energy converted to chemical energy by producers

Question 2

(2 marks)

What prevents the stomach from being digested by its own secretions?

- A. The stomach has a high permeability to hydrogen ions
- B. The presence of beneficial bacteria that neutralize stomach acid
- C. The stomach lining is replaced every month
- D. The stomach lining secretes digestive enzymes only when food is present
- E. The production of bile in the stomach neutralizes the digestive acids

Question 3

(2 marks)

In modern terminology, diversity is understood to be a result of genetic variation. Sources of variation for evolution include all of the following except

A. mistakes in translation of structural genes.

B. mistakes in DNA replication.

C. translocations and mistakes in meiosis.

D. recombination at fertilization.

Question 4

(2 marks)

Charles Darwin was the first person to propose

A. that humans evolved from apes.

B. that evolution occurs.

C. that evolution occurs through a process of gradual change.

D. that the Earth is older than a few thousand years.

E. a mechanism for how evolution occurs.

Question 5

(2 marks)

How can you determine whether or not a given population is in genetic equilibrium?

A. Compare gene frequencies and genotype frequencies of two generations.

B. Evaluate the population's response to environmental stress

C. Assess the mutation rates of the population's genes

D. Compare the population size over generations

E. Calculate the ratio of males to females in the population

True / False

Question 1

(2 marks)

If messenger RNA molecules were not destroyed, they would replicate rapidly, leading to excessive

polypeptide production.

polypeptide production.

True

False

Question 2

(2 marks)

Autoradiography can effectively demonstrate the incorporation of amino acids into proteins within cells.

True

False

Question 3

(2 marks)

Normal somatic cells of a human male do not contain an inactivated X chromosome due to the presence of only one X chromosome.

True

False

Question 4

(2 marks)

Viruses are not independent organisms and require a host cell to replicate.

True

False

Question 5

(2 marks)

The symplastic pathway for sucrose movement includes fibers, bundle sheath, companion cell, and phloem parenchyma.

True

False

Long Form

Question 1

(10 marks)

Discuss the emergence of the first mammals during the Carboniferous Period and analyze the key differences between monotremes and marsupials in terms of their reproductive strategies and adaptations.

Type your answer here...

Question 2

(10 marks)

Discuss the challenges associated with classifying the tissues of higher plants, highlighting the role of evolutionary heritage in their characterization.

Type your answer here...

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